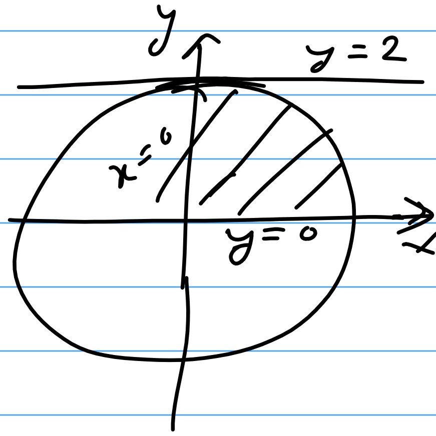
Calculus II : 2023 Fall

1.a) $\int_0^2 \int_0^{\sqrt{4-y^2}} \cos(x^2+y^2)\, dx\, dy$

Sol^n Region: $0 \le x \le \sqrt{4-y^2}$; $x=0,\ x=\sqrt{4-y^2}$

$0 \le y \le 2$; $y=0,\ y=2$



changing to polar,

$0 \le r \le 2$, $0 \le \theta \le \pi/2$

$$= \int_0^{\frac{\pi}{2}} \int_0^2 \cos(r^2\cos^2\theta + r^2\sin^2\theta)\, r\, dr\, d\theta$$

$$= \int_0^{\frac{\pi}{2}} \int_0^2 \cos(r^2) .\ r\, dr\, d\theta$$

Put, $r^2 = u$

$du = 2r\,dr$

when, $r=0,\ u=0$

$r=2,\ u=4$

$$= \int_0^{\frac{\pi}{2}} \int_0^4 \cos u .\ \frac{du}{2}\, d\theta$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} [\sin u]_0^4\, d\theta$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \sin 4\, d\theta$$

$$= \sin 4 . \frac{\pi}{4} \quad //$$

1.b) $\int_0^1 \int_0^1 \int_0^1 (x^2+y^2+z^2)\, dz\, dy\, dx$

Sol^n,

$$\Rightarrow \int_0^1 \int_0^1 \left(x^2 z + y^2 z + \frac{z^3}{3} \right)_0^1 dy\, dx$$

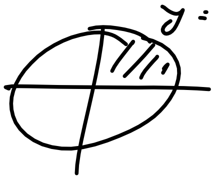
$$\Rightarrow \int_0^1 \int_0^1 \left(x^2 + y^2 + \frac{1}{3} \right) dy\, dx$$

$$\Rightarrow \int_0^1 \int_0^1 \left[x^2 y + \frac{y^3}{3} + \frac{y}{3} \right]_0^1 dx$$

$$\Rightarrow \int_0^1 \left[x^2 + \frac{1}{3} + \frac{1}{3} \right] dx \Rightarrow \int_0^1 \left[x^2 + \frac{2}{3} \right] dx$$

$$\Rightarrow \left[\frac{x^3}{3} + \frac{2x}{3} \right]_0^1 \Rightarrow \left[\frac{1}{3} + \frac{2}{3} \right] \Rightarrow 1 \quad //$$

1.c) $x^2+y^2=4$ and $z+y=3$



Region: $0 \le x \le \sqrt{4-y^2}$
$0 \le y \le 2$

Reqd volume $= \iint_R z\,dA$; $z = 3-y$

$$V = \int_0^2 \int_0^{\sqrt{4-y^2}} (3-y)\,dx\,dy$$

Changing to polar form, $0 \le r \le 2$, $0 \le \theta \le \frac{\pi}{2}$

$$V = \int_0^{\frac{\pi}{2}} \int_0^2 (3 - r\sin\theta)\, r\,dr\,d\theta$$

$$\Rightarrow \int_0^{\frac{\pi}{2}} \int_0^2 (3r - r^2 \sin\theta)\, \underline{dr}\,d\theta$$

$$\Rightarrow \int_0^{\frac{\pi}{2}} \left[\frac{3r^2}{2} - \frac{r^3}{3}\sin\theta \right]_0^2 d\theta$$

$$\Rightarrow \int_0^{\frac{\pi}{2}} \left[6 - \frac{8}{3}\sin\theta \right] d\theta$$

$$\Rightarrow \left[6\theta + \frac{8}{3}\cos\theta \right]_0^{\pi/2}$$

$$\Rightarrow 3\pi + \left(\frac{8}{3}\times 0\right) - 0 - \frac{8}{3}$$

$$\Rightarrow 3\pi - \frac{8}{3} \quad //$$

2.a) Solve using power series, $y'' + (1-x^2)y = 0 \rightarrow (1)$

Soln of (1) is, $y = \sum_{n=0}^{\infty} a_n x^n$

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots \infty \rightarrow (2)$$

Differentiate (2) successively twice,

$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$, $y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$

Put y'' & y in (1),

$$\sum_{n=2}^{\infty} n(n-1)\,a_n x^{n-2} + (1-x^2)\sum_{n=0}^{\infty} a_n x^n = 0$$

Put, $k = n-2$, then replace k with n,

$$\sum_{n=0}^{\infty} (n+2)(n+1)\,a_{n+2}x^n + \sum_{n=0}^{\infty} a_n x^n - \sum_{n=0}^{\infty} a_n x^{n+2} = 0$$

or, $$\sum_{n=0}^{\infty} (n+2)(n+1)\,a_{n+2}x^n + \sum_{n=0}^{\infty} a_n x^n - \sum_{n=2}^{\infty} a_{n-2} x^n = 0$$

Solve for $n=0$ & $n=1$ independently then combine it,

For $n=0$, $2a_2x^0 + a_0x^0 = 0$

$$2a_2 = -a_0$$

$$a_2 = -\frac{a_0}{2}$$

For $n=1$, $6a_3x + a_1x = 0$

$$a_3 = -\frac{a_1}{6}$$

For $n \geq 2$,

$$(n+2)(n+1)a_{n+2} + a_n - a_{n-2} = 0$$

$$a_{n+2} = -\left[\frac{a_n - a_{n-2}}{(n+1)(n+2)}\right]$$

Put $n=2$,

$$a_4 = -\left[\frac{a_2 - a_0}{12}\right] = \frac{-1}{12}\left[\frac{-a_0}{2} - a_0\right] = \frac{a_0}{8}$$

Put $n=3$,

$$a_5 = -\left[\frac{a_3 - a_1}{20}\right] = \frac{-1}{20}\left[\frac{-a_1}{6} - a_1\right] = \frac{7a_1}{120}$$

Put $n=4$.

$$a_6 = -\left[\frac{a_4 - a_2}{30}\right] = \frac{-1}{30}\left[\frac{a_0}{8} + \frac{a_0}{2}\right] = \frac{-a_0}{48}$$

So (2) is,

$$y = a_0 + a_1x - \frac{a_0x^2}{2} - \frac{a_1}{6}x^3 + \frac{a_0}{8}x^4 + \frac{7a_1}{120}x^5 - \frac{a_0}{48}x^6 - \dots$$

$$\therefore y = a_0\left(1 - \frac{x^2}{2} + \frac{x^4}{8} - \frac{x^6}{48} + \dots\right) + a_1\left(x - \frac{x^3}{6} + \frac{7x^5}{120} - \dots\right)$$

Legendre's eqn and it's soln.

= The second order differential eqn of the form,

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0 \rightarrow (1)$$

is known as legendre's differential eqn where n is parameter takes any integer or any real number

Comparing with $y'' + P(x)y' + Q(x)y = 0$

$$P(x) = \frac{-2x}{1-x^2}, \quad Q(x) = \frac{n(n+1)}{1-x^2}$$

Soln of legendre's eqn :

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

$$(1-x^2)y'' - 2xy' + ky = 0 \rightarrow (1), \quad \text{where } k = n(n+1)$$

let,

$$y = \sum_{m=0}^{\infty} a_m x^m \rightarrow (2)$$ be soln of (1),

Differentiating wrt x successfully twice,

$$y' = \sum_{m=0}^{\infty} m a_m x^{m-1}, \quad y'' = \sum_{m=0}^{\infty} m(m-1) a_m x^{m-2}$$

Put y, y', y'' in (1),

or, $(1-x^2)\sum_{m=0} m(m-1)a_m x^{m-2} - 2x\sum_{m=0} m a_m x^{m-1} + k\sum_{m=0} a_m x^m = 0$

or, $\sum_{m=0}^{\infty} m(m-1)a_m x^{m-2} - \sum_{m=0} m(m-1)a_m x^m - 2\sum_{m=0} m a_m x^m + k\sum_{m=0}^{\infty} a_m x^m = 0$

Replace $m = m-2$ in 1st series,

or, $\sum_{m=-2}^{\infty} (m+2)(m+1)a_{m+2} x^m - \sum_{m=0}^{\infty} m(m-1)a_m x^m - 2\sum_{m=0} m a_m x^m + k\sum_{m=0} a_m x^m = 0$

Or, $\sum_{m=0}^{\infty} (m+2)(m+1)a_{m+2} x^m = \sum_{m=0}^{\infty} [m(m-1) + 2m \quad k] a_m x^m$

or, $(m+2)(m+1)a_{m+2} = (m^2 - m + 2m - n^2 - n)a_m$

or, $(m+1)(m+2)a_{m+2} = (m^2 + m - n^2 - n)a_m$

or, $(m+1)(m+2)a_{m+2} = [(m^2 - n^2) + (m+n)]a_m$

or, $(m+1)(m+2)a_{m+2} = (m-n)(m+n+1)a_m$

$$a_{m+2} = \frac{-(n-m)(m+n+1)a_m}{(m+1)(m+2)} \quad \text{for } m \geq 2$$

which is recursive formula for determining a_m for $m \geq 2$ in terms of a_0, a_1 taking $m = 0, 1, 2, 3, \ldots$

Put $m=0$,

$$a_2 = -\frac{(n)(n+1)a_0}{1 \cdot 2}$$

$$a_2 = -\frac{n(n+1)a_0}{2!}$$

Put $m=1$,

$$a_3 = -\frac{(n-1)(n+2)a_1}{2 \cdot 3}$$

$$= -\frac{(n-1)(n+2)a_1}{3!}$$

Put $m=2$,

$$a_4 = -\frac{(n-2)(n+3)a_2}{3 \cdot 4}$$

$$= \frac{n(n+1)(n-2)(n+3)a_0}{4!}$$

Put $m=3$,

$$a_5 = -\frac{(n-3)(n+4)a_3}{4 \cdot 5}$$

$$= \frac{(n-1)(n+2)(n-3)(n+4)a_1}{5!}$$

Substitute coefficients in (2), we get general sol^n of legendre eq^n.

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^4 + a_5 x^5 + \ldots$$

$$y = a_0\left(1 - \frac{n(n+1)x^2}{2!} + \frac{n(n+1)(n-2)(n+3)x^4}{4!} - \ldots\right) + a_1\left(x - \frac{(n-1)(n+2)x^3}{3!} + \frac{(n-1)(n+2)(n-3)(n+4)x^5}{5!} - \ldots\right)$$

which is in the form of

$$y = a_0 y_1(x) + a_1 y_2(x)$$

where $y_1(x) = \ldots \qquad y_2(x) = \ldots$

The $y_1(x)$ and $y_2(x)$ are independent sol^n of legendre's eq^n

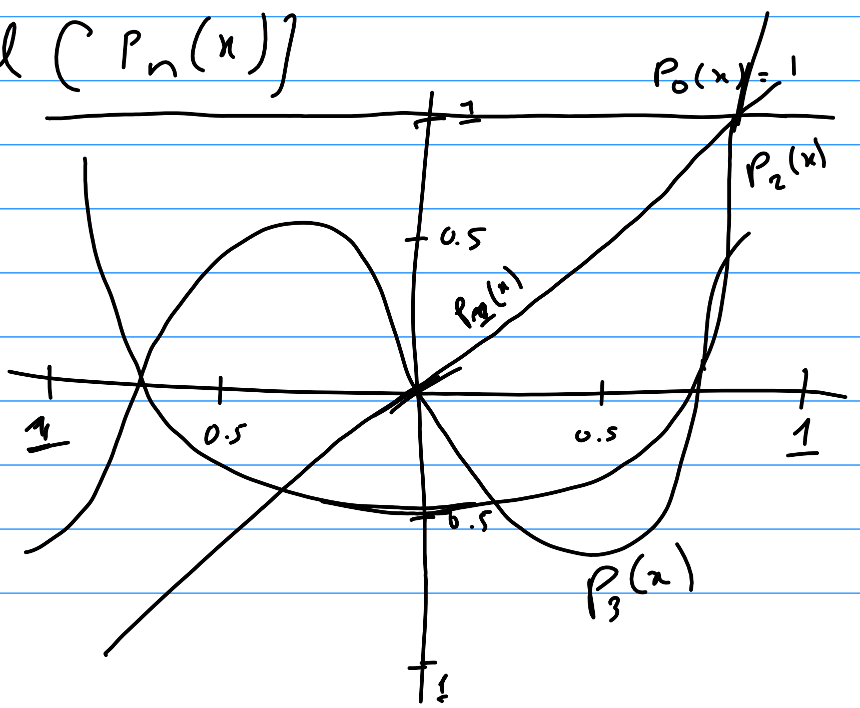
legendre's polynomial $[P_n(x)]$

$P_0(x) = 1$,

$P_1(x) = x$,

$P_2(x) = \frac{1}{2}(3x^2 - 1)$

$P_3(x) = \frac{1}{2}(5x^3 - 3x)$



3.a) 1^{st} shifting laplace transform.

If $L(f(t)) = F(s)$ then,

$$L(e^{at}.f(t)) = [F(s)]_{s \to s-a}$$

i) Find laplace transform of $te^{-t}\cosh t$.

By 1^{st} shifting theorem, $f(t) = e^{-t}\cosh t$

$$F(s) = \left[\frac{s}{s^2-1}\right]_{s \to s+1}$$

$$= \frac{s+1}{(s+1)^2-1} = \frac{s+1}{s^2+2s}$$

Now,

$$L(t.e^{-t}\cosh t) = (-1)^1 \frac{d\,F(s)}{ds} \Rightarrow -\frac{d}{ds}\left(\frac{s+1}{s^2+2s}\right)$$

$$\Rightarrow -\left[\frac{(s^2+2s)-(s+1)(2s+2)}{(s^2+2s)^2}\right]$$

$$\Rightarrow -\left[\frac{s^2+2s-(2s^2+2s+2s+2)}{(s^2+2s)^2}\right]$$

$$\Rightarrow -\left[\frac{-s^2-2s-2}{(s^2+2s)^2}\right]$$

$$\Rightarrow \frac{s^2+2s+2}{(s^2+2s)^2} \quad //$$

ii) Find inverse laplace of $\frac{1}{s^2-5s+6}$

$$f(t) = L^{-1}\left[\frac{1}{s^2-5s+6}\right] \Rightarrow L^{-1}\left[\frac{1}{(s-3)(s-2)}\right]$$

$$= L^{-1}\left[\frac{1}{s^2-5s+\left(\frac{5}{2}\right)^2-\left(\frac{5}{2}\right)^2+6}\right]$$

$$= L^{-1}\left[\frac{1}{\left(s-\frac{5}{2}\right)^2-\left(\frac{1}{2}\right)^2}\right]$$

$$= 2L^{-1}\left[\frac{1/2}{s^2-\left(\frac{1}{2}\right)^2}\right]_{s \to s-\frac{5}{2}}$$

$$= 2e^{\frac{5}{2}.t} \sinh\frac{t}{2}$$

3.b) $y''+4y'+3y = e^{-t}$, $y(0)=y'(0)=1$

Solve,

$$L(y'')+4L(y')+3L(y) = L(e^{-t})$$

or, $[s^2L(y) - s.y(0) - y'(0)] + 4[sL(y)-y(0)] + 3L(y) = \frac{1}{s+1}$

or, $L(y)[s^2+4s+3] - s - 1 - 4 = \frac{1}{s+1}$

or, $L(y)[s^2+4s+3] = \frac{1}{s+1} + s + 5 \Rightarrow \frac{1+s^2+s+5s+5}{s+1}$

or, $L(y)[(s+1)(s+3)] = \frac{s^2+6s+6}{s+1}$

or, $y = L^{-1}\left[\frac{s^2+6s+6}{(s+1)^2(s+3)}\right]$

Now,

$$\frac{s^2+6s+6}{(s+1)^2(s+3)} = \frac{A}{(s+1)} + \frac{B}{(s+1)^2} + \frac{C}{(s+3)}$$

$$s^2+6s+6 = A(s+1)(s+3) + B(s+3) + C(s+1)^2$$

or, Put, $s=-1$,	Put $s=-3$,	Put $s=0$
or, $1 = 2B$	$-3 = 4c$	$6 = 3A + 3/2 - 3/4$
$\therefore B = 1/2$	$\therefore c = -3/4$	$\therefore A = 7/4$

Now,

$$y = L^{-1}\left[\frac{7}{4(s+1)} + \frac{1}{2(s+1)^2} - \frac{3}{4(s+3)}\right]$$

$$y = \frac{7}{4}e^{-t} + \frac{1}{2}e^{-t}.t - \frac{3}{4}e^{-3t}$$

4.a) $\phi = \log(x^2+y^2+z^2)$, find div(grad ϕ)

Soln

$$\text{grad.}\phi = \vec{i}\frac{\partial \phi}{\partial x} + \vec{j}\frac{\partial \phi}{\partial y} + \vec{k}\frac{\partial \phi}{\partial z}$$

$$= \vec{i}\,\frac{2x}{x^2+y^2+z^2} + \vec{j}\,\frac{2y}{x^2+y^2+z^2} + \vec{k}\,\frac{2z}{x^2+y^2+z^2}$$

let $\vec{P}$ = grad ϕ

$$\text{div}(\text{grad}\,\phi) = \nabla\cdot\vec{P} = \frac{\partial P_1}{\partial x} + \frac{\partial P_2}{\partial y} + \frac{\partial P_3}{\partial z}$$

$$= \frac{2(x^2+y^2+z^2)-2x(2x)}{(x^2+y^2+z^2)^2} + \frac{2(x^2+y^2+z^2)-2y\cdot 2y}{(x^2+y^2+z^2)^2} + \frac{2(x^2+y^2+z^2) - 2z\cdot 2z}{(x^2+y^2+z^2)^2}$$

$$= \frac{2(y^2+z^2-x^2)+2(x^2+z^2-y^2)+2(x^2+y^2-z^2)}{(x^2+y^2+z^2)^2}$$

$$= \frac{2}{(x^2+y^2+z^2)} \quad //$$

4.b) $D_{\vec{a}}(f) = ?$, $f = x^2yz + 4xz^2$ at $(1,-2,1)$ in direction $2\vec{i} - \vec{j} - 2\vec{k}$

Soln

$$D_{\vec{a}}(f) = \nabla\vec{f}\cdot\frac{\vec{a}}{|\vec{a}|}$$

Here, $\nabla\vec{f} = \vec{i}\frac{\partial f}{\partial x} + \vec{j}\frac{\partial f}{\partial y} + \vec{k}\frac{\partial f}{\partial z} = \vec{i}(2xyz+4z^2) + \vec{j}(x^2z) + \vec{k}(x^2y+8xz)$

at $(1,-2,1)$

$$\nabla\vec{f} = 0\cdot\vec{i} + \vec{j} + 6\vec{k}$$

let $\vec{a} = 2\vec{i} - \vec{j} - 2\vec{k}$

$$\therefore D_{\vec{a}}(f) = (\vec{j}+6\vec{k})\cdot\frac{(2\vec{i}-\vec{j}-2\vec{k})}{\sqrt{2^2+1^2+2^2}}$$

$$= \frac{-1-12}{3} = -\frac{13}{3} \quad //$$

4.c) $\oint_C \vec{F} \cdot d\vec{r} = ?$, $\vec{F} = (y, z, x)$, $C: \vec{r} = (t, t^2, t^3)$

from $(0, 0, 0)$ to $(2, 4, 8)$

Sol^n) $\vec{F} = y\vec{i} + z\vec{j} + x\vec{k}$

$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$,

$d\vec{r} = dx\,\vec{i} + dy\,\vec{j} + dz\,\vec{k}$

let $x = t$, $\quad y = t^2 \quad z = t^3$

$dx = dt \quad dy = 2t\,dt \quad dz = 3t^2$

$\int_C \vec{F}\, d\vec{r} = \int_C (y\,dx + z\,dy + x\,dz)$,

let $x = t$
where $x = 0, t = 0$
$x = 2, t = 2$

$= \int_0^2 \left(t^2 dt + t^3 . 2t \; dt + t . 3t^2 \right)$

$= \left[\frac{t^3}{3} + 2\frac{t^5}{5} + 3\frac{t^4}{4} \right]_0^2$

$= \frac{412}{15}$ //

5. a) greens theorem:

→ Let R be the closed bounded region on xy plane bounded by curve. Let $F_1(x, y)$ and $F_2(x, y)$ be two functions that are continuous and their first order partial derivatives are also continuous on R. Then,

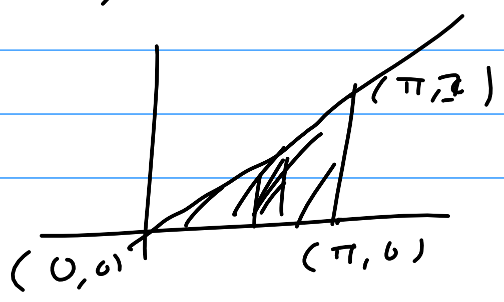
$$\oint_C (F_1 dx + F_2 dy) = \iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dx\,dy$$

$\vec{F} = (\sin y, \cos x) = \sin y\,\vec{i} + \cos x\,\vec{j}$, $C: (0, 0), (\pi, 0), (\pi, 1)$

Sol^n

$\vec{F} = \sin y\,\vec{i} + \cos x\,\vec{j}$

$\frac{\partial F_1}{\partial y} = \cos y$, $\frac{\partial F_2}{\partial x} = -\sin x$



Eqn of st line from $(0,0)$ & $(\pi, 1)$

$$y - 0 = \frac{1-0}{\pi - 0}(x - 0)$$

$$y = \frac{x}{\pi}$$

Using green theorem,

$$\iint_R (-\sin x - \cos y)\, dy\, dx \Rightarrow \int_0^{\pi}\int_0^{x/\pi} -(\sin x + \cos y)\, dy\, dx$$

$$\Rightarrow -\int_0^{\pi} \left[y \cdot \sin x + \sin y\right]_0^{x/\pi} dx$$

$$\Rightarrow -\int_0^{\pi} \left[\frac{x}{\pi}\sin x + \sin\frac{x}{\pi}\right] dx$$

$$\Rightarrow -\left[-\frac{x}{\pi}\cos x + \frac{\sin x}{\pi} - \pi \frac{\cos x}{\pi}\right]_0^{\pi}$$

$$\Rightarrow \left[\frac{x}{\pi}\cos x - \frac{\sin x}{\pi} + \pi \frac{\cos x}{\pi}\right]_0^{\pi}$$

$$\Rightarrow \left[\frac{-\cancel{\pi}}{\cancel{\pi}} - 0 + \pi \cos 1 - (0 - 0 + \pi)\right]$$

$$\Rightarrow \pi \cos 1 - \pi - 1$$

5.b) $\oint_C \vec{F} \cdot \vec{dr} = ?$ using stoke's theorem,

$\vec{F} = y^2 \vec{i} + z^2 \vec{j} + x^2 \vec{k}$ $\quad$ $S: x + y + z = 1$ in 1^{st} octant

Soln

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y^2 & z^2 & x^2 \end{vmatrix} = \vec{i}(0 - 2z) - \vec{j}(2x - 0) + \vec{k}(0 - 2y)$$

$$= -2z\vec{i} - 2x\vec{j} - 2y\vec{k}$$

Let $\vec{r}$ be position vector,

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k} = x\vec{i} + y\vec{j} + (1 - x - y)\vec{k}$$

$$\vec{r_x} = \vec{i} + 0 \cdot \vec{j} - \vec{k}, \quad \vec{r_y} = 0 \cdot \vec{i} + \vec{j} - \vec{k}$$

$$\vec{N} = \vec{r_x} \times \vec{r_y} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{vmatrix} = +\vec{i} + \vec{j} + \vec{k}$$

$$(\nabla \times \vec{F}).\vec{N} = -2z - 2x - 2y$$
$$= -2(1-x-y) - 2x - 2y$$
$$= -2 + 2x + 2y - 2x - 2y$$
$$= -2$$

Region: Take projection on xy plane, $z = 0$

$$y = 1 - x$$

Put $y = 0$, $x = 1$

By stroke theorem,

$$\oint \vec{F}.d\vec{r} = \int_0^1 \int_0^{1-x} -2 \, dy \, dx = \int_0^1 (-2)[y]_0^{1-x} dx$$

$$\Rightarrow -2 \int_0^1 (1-x) \, dx$$
$$\Rightarrow -2 \left[x - \frac{x^2}{2} \right]_0^1 \Rightarrow -2\left[1 - \frac{1}{2}\right] = -1$$

OR,

$\iint_S \vec{F} \vec{n} \, dA = ?$ surface integral = ?

$\vec{F} = x^2\vec{i} + y^2\vec{j} + z^2\vec{k}$. $\quad \vec{r} = (u\cos v, u\sin v, 3v)$.

$0 \le u \le 1$, $0 \le v \le 2\pi$

Soln,

$$\vec{r} = u\cos v\,\vec{i} + u\sin v\,\vec{j} + 3v\vec{k}$$
$$\vec{r_u} = \cos v\,\vec{i} + \sin v\,\vec{j} + 0.\vec{k}, \quad \vec{r_v} = -u\sin v\,\vec{i} + u\cos v\,\vec{j} + 3\vec{k}$$

$$\vec{N} = \vec{r_u} \times \vec{r_v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \cos v & \sin v & 0 \\ -u\sin v & u\cos v & 3 \end{vmatrix} = \vec{i}(3\sin v) - \vec{j}(3\cos v) + \vec{k}(u\cos^2 v + u\sin^2 v)$$
$$= 3\sin v.\vec{i} - 3\cos v\vec{j} + u\vec{k}$$

$x = u\cos v$, $y = u\sin v$, $z = 3v$

$$\vec{F}.\vec{N} = 3u^2\cos^2 v\sin v - 3u^2\sin^2 v.\cos v + 9uv^2$$

$$\int_0^{2\pi}\int_0^1 (3u^2\cos^2 v\sin v - 3u^2\sin^2 v\cos v + 9uv^2)\,du\,dv$$

$$\Rightarrow \int_0^{2\pi} \left[u^3\cos^2 v\sin v - u^3\sin^2 v.\cos v + \frac{9u^2v^2}{2} \right]_0^1$$

$$\Rightarrow \int_0^{2\pi} \left[\cos^2 v\sin v - \sin^2 v\cos v + \frac{9v^2}{2} \right] dv$$

$\Rightarrow$ { Integrals of $\cos^2 v \sin v$, $\sin^2 v . \cos v$ over $[0, 2\pi]$ are 0 due to periodicity.

$$= \frac{9}{2}\int_0^{2\pi} v^2 \, dv = \frac{9}{2}\left[\frac{v^3}{3}\right]_0^{2\pi} = \frac{9}{2} \times \frac{8\pi^3}{3} = 12\pi^3$$

6. a) $f(x) = x + |x|$, $-\pi < x < \pi$

Sol^n,

Period $= 2\pi$

$$|x| = \begin{cases} x & \text{if } x > 0 \\ -x & \text{if } x < 0 \end{cases}$$

$$x + |x| = \begin{cases} 2x & \text{if } 0 < x < \pi \\ 0 & \text{if } -\pi < x < 0 \end{cases} \Rightarrow \begin{cases} 0 & -\pi < x < 0 \\ 2\pi & 0 < x < \pi \end{cases}$$

The fourier series of $f(x)$ in $(-\pi, \pi)$ is,

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

where,

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} (x + |x|)\,dx = \frac{1}{2\pi}\left[\int_{-\pi}^{0} f(x)\,dx + \int_0^{\pi} f(x)\,dx\right]$$

$$= \frac{1}{2\pi}\int_0^{\pi} 2x \, dx$$

$$= \frac{1}{\pi}\left[\frac{x^2}{2}\right]_0^{\pi}$$

$$= \frac{\pi}{2}$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx \, dx$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0} f(x)\cos nx\,dx + \int_0^{\pi} f(x)\cos nx\,dx\right]$$

$$= \frac{1}{\pi}\int_0^{\pi} 2x \cos nx \, dx$$

$$= \frac{2}{\pi}\left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2}\right]_0^{\pi} = \frac{2}{\pi}\left[\pi \times 0 + \frac{(-1)^n}{n^2} - \frac{1}{n^2}\right]$$

$$= \frac{2}{\pi n^2}\left((-1)^n - 1\right)$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0} f(x)\sin nx\,dx + \int_{0}^{\pi} f(x)\sin nx\,dx\right]$$

$$= \frac{1}{\pi}\int_{0}^{\pi} 2x\sin nx\,dx$$

$$= \frac{2}{\pi}\left[-\frac{x\cos nx}{n} + \frac{\sin nx}{n^2}\right]_0^{\pi} \Rightarrow \frac{2}{\pi}\left[\frac{-\pi(-1)^n}{n}\right]$$

$$= \frac{-2(-1)^n}{n}$$

Now fourier series is,

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty}\left(\frac{2}{\pi n^2}\left[(-1)^n - 1\right]\cos nx - \frac{2}{n}(-1)^n \sin nx\right) //$$

6.b) $f(x) = x^2$, $0 < x < \pi$, half range fourier cosine series and deduce that

$$\frac{\pi^2}{6} = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots$$

soln

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx \longrightarrow (1)$$

$$a_0 = \frac{1}{\pi}\int_0^{\pi} f(x)\,dx = \frac{1}{\pi}\left(\frac{x^3}{3}\right)_0^{\pi} = \frac{\pi^2}{3}$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx = \frac{2}{\pi}\int_0^{\pi} x^2\cos nx\,dx$$

$$= \frac{2}{\pi}\left[x^2\frac{\sin nx}{n} - \int \frac{2x\sin nx\,dx}{n}\right]$$

$$= \frac{2}{\pi}\left[\frac{x^2\sin nx}{n} - \frac{2}{n}\left[-\frac{x\cos nx}{n} + \frac{\sin nx}{n^2}\right]\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[\frac{x^2\sin nx}{n} + \frac{2x\cos nx}{n^2} - \frac{2\sin nx}{n^3}\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[\cancel{\frac{\pi^2}{n}}\,0 + \frac{2\pi(-1)^n}{n^2} - 0 - (0 + 0 - 0)\right]$$

$$= \frac{4}{n^2}(-1)^n$$

Now,

$$f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2}(-1)^n \cos nx$$

$$x^2 = \frac{\pi^2}{3} + 4\left[-\frac{\cos x}{1^2} + \frac{\cos 2x}{2^2} - \frac{\cos 3x}{3^2} + \frac{\cos 4x}{4^2} - \cdots\right]$$

$$x^2 = \frac{\pi^2}{3} - 4\left[\cos x - \frac{\cos 2x}{4} + \frac{\cos 3x}{9} - \frac{\cos 4x}{16} + \cdots\right]$$

Put $x = \pi$

$$\pi^2 = \frac{\pi^2}{3} - 4\left[-1 - \frac{1}{4} - \frac{1}{9} - \frac{1}{16} - \cdots\right]$$

$$\frac{2\pi^2}{3} = 4\left[1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots\right]$$

$$\frac{\pi^2}{6} = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots$$

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